Abhishek Chettiar

AI/ML Engineer | Machine Learning, Generative AI & Python

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PROFESSIONAL SUMMARY

AI/ML Engineer with hands-on experience developing machine learning and computer vision models and building LLM-powered applications. Skilled in Python, feature engineering, model evaluation and explainability, with experience delivering document-intelligence pipelines, retrieval-augmented generation (RAG) assistants and workflow automation using Grok, Ollama, FastAPI/Flask and Docker.

PROFESSIONAL EXPERIENCE

TechBerry Infotech Pvt. Ltd. Nov 2025 – Present

Junior AI/ML Engineer *(previously AI/ML Trainee)*

- Build Python AI applications for recruitment, ticketing, workforce and internal-support products, integrating **Grok and other LLM APIs** for document understanding, classification and structured extraction.
- Extended hosted-LLM workflows to support **self-hosted inference through Ollama**, with task-specific prompts, structured extraction and provider fallbacks.
- Contribute shared model-provider interfaces, output validation, background jobs and retries across projects.
- Integrate authentication, OTP and notifications; deploy and troubleshoot via Docker, GitHub Actions and n8n.

WAI Technologies Apr 2025 – Jul 2025

AI/ML Software Developer Intern

- Trained a YOLOv11 detector on 10,000+ annotated images, achieving **93% precision** for real-time inference.
- Built AI-workflow APIs and automated deployment/configuration with n8n, reducing manual effort by about **35%**.

TECHNICAL SKILLS

Machine Learning: scikit-learn, XGBoost, LightGBM, CatBoost, Isolation Forest, LSTM, ARIMA; feature engineering, stratified validation, model evaluation, SHAP/LIME explainability

Deep Learning & Computer Vision: PyTorch, TensorFlow/Keras, YOLOv5/YOLOv11, ResNet50, OpenCV; transfer learning, data augmentation

LLMs & Generative AI: Grok, GPT-4.1 and Claude APIs; Ollama local inference (Llama, Qwen); prompt engineering, structured outputs, Pydantic/schema validation, LangChain, Hugging Face Transformers

Document AI & Retrieval: OCR and NLP document processing, chunking, embeddings, semantic search, ChromaDB, FAISS, BM25 with RRF, cross-encoder reranking, reference-based RAG evaluation

Python & Data: Python, SQL, Pandas, NumPy, PostgreSQL, SQL Server, SQLAlchemy

Backend & Pipelines: Flask, FastAPI, REST APIs, Redis, Celery, batch processing, background workers

Deployment & Orchestration: Docker, Git, GitHub Actions CI/CD, Linux/systemd, GCP, n8n, Streamlit, Gradio

PROJECTS

HR Intelligence Platform | *Flask, PostgreSQL, Grok, Ollama*

- Built **AI-assisted résumé and job-description parsing** with Grok and locally hosted Ollama models, combining text extraction/OCR and deterministic parsing with schema-validated LLM enrichment and caching.
- Built **bulk résumé processing** with chunked/ZIP uploads, concurrent workers, controlled model calls, validation, retries, progress tracking, pause/resume and exports.
- Turned structured candidate profiles and job requirements into **explainable matching scores** with matched/missing-requirement summaries.

RAG Knowledge Assistants — Employee Assist & Company Knowledge | *Ollama, Qwen, ChromaDB*

- Built Employee Assist, an HR assistant in Microsoft Teams, and a company-knowledge chatbot on a shared local stack: FAQ-aware ingestion, embeddings, ChromaDB retrieval and Qwen inference through Ollama.
- Added topic routing, answer caching and **retrieval-confidence gates** to limit model work on a CPU-only VM, with background model warm-up and thread offloading for responsive request handling.
- Implemented configurable dense/BM25 retrieval with reciprocal rank fusion and optional cross-encoder reranking.
- For the company-knowledge chatbot, added **reference-based evaluation** using paraphrased knowledge-base questions, rubric scoring and evaluation-history exports.

Credit Default Risk Prediction | *XGBoost, LightGBM, CatBoost, SHAP, FastAPI*

- Built a credit-risk pipeline with feature engineering, **leakage-safe preprocessing** and stratified validation, serving SHAP-explained predictions through a Docker-packaged FastAPI app.

Biometric Workforce Intelligence | *FastAPI, Celery, Redis, Grok*

- Co-developed device synchronization and attendance-session reconstruction for ZKTeco/ESSL terminals, with safeguards against duplicate and stale punch records.
- Combined rule-based attendance alerts with **Isolation Forest anomaly detection** over six daily workforce features and heuristic fallback.
- Implemented Grok management and payroll insights with validated JSON, caching, retries and a circuit breaker.

American Sign Language Detection | *YOLOv5, PyTorch, OpenCV, Gradio*

- Trained YOLOv5 on 10,000+ annotated images with augmentation and transfer learning, achieving **91% mAP@0.5**; built live webcam inference with a Gradio interface.

TasQBerry Ticket and Workflow Management | *FastAPI, PostgreSQL, Redis, Grok*

- Co-developed **Grok-powered email understanding** that classifies tasks and extracts structured ticket fields from emails and attachments, guided by saved human feedback, with fallback handling.
- Linked AI interpretation to workflow execution: ticket creation, reply association, impact/urgency scoring, SLA reminders and **scheduled recurring-ticket generation**.
- Added duplicate-prevention safeguards, durable outboxes, retries and dead-letter handling for background jobs.

DataVitals Database Observability Copilot | *FastAPI, Ollama*

- Co-developed monitoring for PostgreSQL, MySQL, SQL Server, Oracle and MongoDB, with alert grouping, repeat suppression and multi-channel delivery.
- Correlated health, metric and log signals with YAML runbooks for **evidence-grounded copilot answers**, summarized by configurable Ollama models with deterministic fallbacks.

Agentic Design to Code Assistant | *GPT-4.1, Claude, n8n*

- Orchestrated GPT-4.1 workflows with Claude fallback in n8n to turn design prompts into code, linked to Codespaces and CodeSandbox previews, cutting design-to-deployment time by **40%**.

Explainable Heart Disease Prediction | XGBoost classifier at 84% accuracy with SHAP/LIME explanations.

Human Activity Recognition | LSTM and Random Forest on sensor features; F1 score of 0.89 (six activities).

Dog Breed Classification | ResNet50 transfer learning across 120+ breeds with 92.1% accuracy.

Stock Price Forecasting | LSTM/ARIMA forecasts with 22% lower MSE than a linear-regression baseline.

DB AutoPro Database Automation | Co-developed backup/restore automation across five database engines, PostgreSQL migration workflows and unified operational dataset exports.

Asset Tracking and Lifecycle Management | Co-developed lifecycle tracking with append-only history.

Learning Management System | Co-developed role-based portals with assessments and payments.

Sahakar Bharati Participant Platform | Co-developed Android/web registration and profile workflows with OTP verification, authenticated sessions and FastAPI/PostgreSQL services.

EDUCATION

PG Diploma in Artificial Intelligence and Machine Learning | MIT World Peace University, Pune 2025

Bachelor of Engineering in Mechanical Engineering | A. P. Shah Institute of Technology, Thane 2023

CERTIFICATIONS & ADDITIONAL INFORMATION

Certifications: Machine Learning A-Z; Artificial Intelligence A-Z; Python Deep Learning for Beginners; Data Science Course: Complete Bootcamp; Python Programming Masterclass

Leadership: Treasurer, IEEE APSIT; Head of Design and Body Works, ARC Club, APSIT

Languages: English, Tamil and Hindi (fluent); Marathi (intermediate)